

Yes — that is a fair reading, with one sharpening so you do not credit the reduction with more than it delivers.

What is solid is the chain that makes the reduction legitimate: deformation is a shift, so

$$\mu_n^{(c)} = \text{dist}(-c, \text{spec } \Phi_n).$$

That genuinely converts “why does the plotted number oscillate in n ?” into “how does the root cloud $\text{spec } \Phi_n$ — in particular its inner edge near $-c$ — move as n advances?” So the oscillation is a fact about the geometry of the clouds, one n at a time, and that part is exact, not heuristic.

But notice what kind of reduction it is. It is a *restatement*, not an explanation. I have rewritten the quantity $\mu_n^{(c)}$ as a geometric distance, which relocates the mystery from “the plot” to “the inner edge of $\text{spec } \Phi_n$ ” — but I have not said why that inner edge moves with a period-4 rhythm, why the nearest eigenvalue is real, or what sets the envelope. Those are exactly the things left open in the note. A reduction earns its keep when the target side is more tractable than the source side; here the target (the zeros of the Appell polynomials Φ_n attached to $G = \exp \int J$) is better-structured than a raw plot, but I have not actually solved or even controlled it for τ . So “reduced to the geometry of the root cloud” is accurate; “explained via the geometry of the root cloud” would be an overclaim.

There is also a layer below the cloud geometry that the phrase hides. The clouds $\text{spec } \Phi_n$ are not free-standing — they are the zero sets of a single sequence of polynomials whose coefficients are read off from one analytic object, $G(t) = \exp \int_0^t J$. The cloud-geometry picture is the *intermediate* level; the *generating* level is G and its boundary singularities. So if you want a chain it runs

oscillation of $\mu_n^{(c)} \iff$ motion of the inner edge of $\text{spec } \Phi_n \iff$ analytic structure of G .

I established the first link exactly, observed the middle one computationally at small n , and only gestured at the third. The genuine reduction I would stand behind is to the analytic structure of G ; the cloud geometry is where that structure becomes visible, not where it originates.

One caveat worth keeping attached: the cloud-geometry statement is exact as mathematics, but my *evidence* for the specific anatomy (real nearest root, period 4) is computational over $n \leq 44$. So the reduction is rigorous; the description of what it reduces to is, so far, empirical.